

NEPHROLOGY TEACHING SERIES

Kidney Transplant

Outpatient management of the transplant recipient

Every visit: graft function, immunosuppression, infection, malignancy, CV risk.

Dr. Cheng • Premier Nephrology Medical Group

MS3/MS4 rotation teaching

Why we get consulted

Transplant patients are more fragile than they look. Small changes matter.

CLINIC VISIT PRIORITIES

Graft function (Cr, UACR, BP, weight)

Tacrolimus trough

Infection screen: BK, CMV, EBV; PJP prophylaxis

Cancer: skin, PTLD, cervix, colon

CV risk, DM, HTN, lipids

CNI + steroid bone effects

NO live vaccines on immunosuppression

TEACHING PEARL

*A rising Cr in transplant is never 'mild.'
Ddx: dehydration, drug interaction, BK,
rejection, CNI toxicity, recurrence,
obstruction. Get tac level, UACR, UA.*

Standard immunosuppression

Induction → maintenance triple therapy → lifelong monitoring.

Phase	Agent	Dose / target	Key toxicity
Induction	Basiliximab (low-risk) or anti-thymocyte globulin (high-risk)	IV, peri-op	ATG: cytokine release, cytopenias, infection
Maintenance	Tacrolimus (CNI)	Trough ~9–12 ng/mL month 1 → 5–7 after yr 1 (center-specific)	Nephrotoxicity, DM, tremor, HTN, hypoMg
	Mycophenolate (mycophenolate mofetil (MMF))	1 g PO BID	GI (diarrhea), cytopenias, teratogen
	Prednisone (if not steroid-free)	5 mg/day maintenance	DM, osteoporosis, cataracts, weight gain
Alternative	Belatacept (CD80/86 blockade)	IV infusion monthly	Only in EBV-seropositive; PTLD risk in naïve

Rising creatinine — the differential

Biopsy is the definitive answer when pre-renal and drug causes are excluded.

PRE-RENAL / HEMODYNAMIC

Dehydration (GI illness, poor PO), diuretics, NSAIDs, ACEi/ARB addition, sepsis

DRUG / TOXIC

High tac level (check trough); cytochrome P450 3A4 (CYP3A4) interactions (azoles, macrolides, diltiazem, grapefruit); contrast; aminoglycosides

REJECTION (need biopsy)

Cellular rejection: T-cell mediated (T-cell infiltrate). Antibody-mediated: donor-specific antibody (DSA)+, C4d+ on biopsy. Both cause rapid Cr rise, often subclinical

INFECTION / VIRAL

BK nephropathy (check BK PCR; decoy cells); CMV; pyelonephritis (allograft tender)

STRUCTURAL

Ureteral obstruction (stricture, lymphocele, stone); renal artery stenosis (refractory HTN + Cr rise); DVT/RVT

RECURRENT DISEASE

IgA (most common on biopsy), FSGS (can recur within days; most graft loss), C3G/MPGN, diabetic nephropathy (years)

Infection prophylaxis and screening

Highest risk first 6 months. Keep surveillance lifelong.

Pathogen	Risk window	Prophylaxis / screening
Pneumocystis (PJP)	First 6–12 mo	TMP-SMX single-strength daily or 3×/wk
CMV	First 3–6 mo (D+/R– highest risk)	Valganciclovir 900 mg daily × 6 mo (D+/R–); or preemptive with PCR monitoring
BK virus	First 2 yr; lifelong surveillance	Screen PCR monthly to month 9, then q3 mo to 2 yr; reduce IS if viremia
EBV / PTLN	Lifelong (high in D+/R–)	PCR if new lymphadenopathy, cytopenia, weight loss
Fungal (Candida, Aspergillus)	Early post-op; lung tx higher risk	Oral nystatin × 3 mo; consider azole if prior col.
HBV reactivation	Lifelong if HBsAg+ or anti-HBc+	Entecavir / tenofovir + serology monitoring
Flu / COVID / S. pneumo	Annual / per schedule	Inactivated vaccines only; NO live (MMR, varicella, yellow fever)

Tacrolimus — levels, interactions, side effects

The single most common source of transplant chaos.

RAISE THE LEVEL (CYP3A4 inhibitors)

Azole antifungals (fluconazole, voriconazole, posaconazole)
Macrolides (erythromycin, clarithromycin — NOT azithromycin)
Diltiazem, verapamil
Protease inhibitors (HIV, HCV)
Grapefruit juice

Expect tac level to rise within 48–72 h; recheck & adjust

LOWER THE LEVEL (CYP3A4 inducers)

Rifampin, rifabutin
Phenytoin, carbamazepine, phenobarbital
St. John's wort

Classic side effects:

Nephrotoxicity (afferent vasoconstriction), hyperK, hypoMg, tremor, HA, new-onset DM, HTN, hair changes, alopecia (in women)

Always draw trough 12 h post-dose

Malignancy — the long-tail risk

Transplant recipients have 2–4× baseline cancer risk. Most cancers are preventable or detectable.

SKIN CANCER

Most common (squamous cell carcinoma (SCC) > basal cell carcinoma (BCC)); 65× baseline for SCC. Annual dermatology. Daily SPF 30+. Self-exam. Avoid tanning beds.

PTLD

EBV-driven B-cell lymphoma; first year risk highest. Symptoms: weight loss, adenopathy, cytopenia. Rx: reduce IS + rituximab ± chemo.

NATIVE KIDNEY RCC

Acquired cystic disease and long dialysis vintage raise RCC risk. Screen high-risk patients by center protocol; routine annual CT is not universal.

GI / CERVICAL / BREAST

Colonoscopy per guidelines; cervical cytology annually (not 3-yearly); mammo per guidelines. HPV vaccination where applicable.

Landmark trials in transplant

What shapes current induction / maintenance / rejection.

Trial	Year	Takeaway
ELITE-Symphony	2007	Low-dose tac + MMF + steroids + daclizumab best — now standard.
BENEFIT	2010	Belatacept vs cyclosporine — better renal function, similar graft survival long-term.
BENEFIT-EXT	2010	Belatacept in extended-criteria donors — non-inferior; avoid in EBV-naïve.
3C Study	2014	Alemtuzumab vs basiliximab induction — equivalent outcomes.
CAESAR	2007	Cyclosporine withdrawal → more rejection; ongoing low-dose CNI is safer.
STAR	2019	Living donor kidney paired exchange — durable benefit, safe.
DeKAF	2012	Antibody-mediated rejection classification and outcomes.
ADVANCE / PIVOT	2012+	Once-daily tacrolimus (Envarsus / Advagraf) — improved adherence, non-inferior.

Common clinic mistakes

The errors that lead to rejection or hospitalization.

AVOID

- Missing drug interactions on tac
- Stopping PJP prophylaxis early
- LIVE vaccines on immunosuppression
- Ignoring small Cr rises
- No routine BK PCR
- Diagnosing rejection without biopsy
- Acting on one tac level

DO INSTEAD

- Review meds at every visit (incl OTC / grapefruit)
- Coordinate prophylaxis with transplant center
- Inactivated vaccines only (recombinant zoster OK)
- Cr ↑ → tac level, BK PCR, UA
- BK + CMV PCR per center schedule
- Biopsy if Cr ↑ > 20% without reversible cause
- Recheck tac 3–5 d after dose change

Clinical case

Think it through — answer is on the next slide.

VIGNETTE

A 55-year-old woman 14 months s/p deceased donor kidney transplant, on tacrolimus 3 mg BID (last trough 6.5), MMF 1 g BID, prednisone 5 mg. Baseline Cr 1.3. Clinic visit today: Cr 1.9. She reports 'taking drugs as prescribed.' New prescription 1 month ago from PCP: clarithromycin for bronchitis, completed. No fever, no urinary symptoms.

QUESTION: What's the most likely cause of the Cr rise, and what should you check?

Clinical case — answer

Key teaching points from the case.

ANSWER

Tacrolimus toxicity from clarithromycin (CYP3A4 inhibition). Check tac trough NOW; expect elevated; also BK PCR, UA, UACR, volume status to exclude other causes.

TEACHING

Drug interactions are the #1 reversible cause of Cr rise in transplant clinic. Classic CYP3A4 inhibitors that RAISE tac: azole antifungals (fluc, vor, posa), macrolides (erythro, clarithro — NOT azithro), diltiazem, verapamil, grapefruit, HIV protease inhibitors. Always review the med list at every visit. If tac level is elevated, reduce dose and re-check in 3-5 days. If Cr doesn't return to baseline, the differential broadens: dehydration, rejection, BK nephropathy, obstruction, recurrent disease. Always get UA, UACR, BK PCR, and consider biopsy if Cr > 20% above baseline without clear reversible cause.

REFERENCES

Kidney Transplant

Guidelines

KDIGO 2009 Care of the Kidney Transplant Recipient

KDIGO 2020 Transplant Candidate Evaluation

AST Infectious Diseases Community of Practice

AST Cancer Screening Consensus

ISHLT vaccines in solid organ transplant (2019)

Landmark Trials

ELITE-Symphony — NEJM 2007

BENEFIT — AJT 2010

BENEFIT-EXT — AJT 2010

3C Study — Lancet 2014

CAESAR — AJT 2007

DeKAF — AJT 2012

STAR (paired exchange) — JAMA 2019

UpToDate (2026 access):

Evaluation of acute kidney allograft dysfunction • Pharmacology and side effects of calcineurin inhibitors • BK polyomavirus-associated nephropathy • Infection after kidney transplantation • Kidney transplantation in adults: Immunosuppression maintenance